

Functional Annotation of *paaJ* (PP_3280, UniProt Q88HS3) in *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

paaJ (locus PP_3280; UniProt Q88HS3; EMBL AAN68887.1) encodes the **3-oxoadipyl-CoA / 3-oxo-5,6-dehydrosuberyl-CoA thiolase** of the aerobic **phenylacetate (phenylacetyl-CoA) catabolic pathway** in *Pseudomonas putida* KT2440. It is a **soluble, cytoplasmic β -ketothiolase** (thiolase-like superfamily; acetyl-CoA C-acyltransferase, EC 2.3.1.174 / EC 2.3.1.223-type) that catalyzes the **thiolytic (retro-Claisen) cleavage of two β -keto-CoA intermediates** in the lower, β -oxidation-like segment of the pathway, releasing **acetyl-CoA** at each step and ultimately producing **succinyl-CoA**, which enters the TCA cycle. It is the terminal carbon-harvesting enzyme of the phenylacetyl-CoA catabolon.

2. Gene / Protein Identity Verification (mandatory check)

The task flagged possible ambiguity because Q88HS3 carries an InterPro "**PcaF**" tag (IPR012793). This was resolved:

- **The InterPro PcaF tag denotes broad thiolase-family membership, not identity with PcaF.** PcaF proper is the **β -ketoadipyl-CoA thiolase of the β -ketoadipate (protocatechuate/catechol) pathway**. Q88HS3 instead carries the diagnostic substrate **3-oxo-5,6-dehydrosuberyl-CoA**, which is unique to the **aerobic phenylacetyl-CoA (paa) degradation pathway**.
- Genomic analysis of KT2440 (Jiménez et al. 2002) explicitly separates the **phenylacetate pathway (pha/paa genes)** from the **β -ketoadipate pathway (pca/cat genes)** [PMID 12534466]. *paaJ*/PP_3280 belongs to the former.
- **Nomenclature note:** the same enzyme appears in the literature under several names across strains — **paaJ** (E. coli-type unified nomenclature and the EMBL annotation used here), **phaD** (the KT2440 "pha" annotation of Jiménez 2002 / Kim 2006), and the *paa*-cluster thiolase in *P. putida* U. All refer to the phenylacetate-pathway thiolase, distinct from PcaF.

Conclusion: The gene symbol is correctly matched; the organism is *P. putida* KT2440; the family/domain content is consistent with a degradative thiolase. Research proceeds on the correct protein.

3. Primary Molecular Function

Reaction catalyzed

PaaJ is a **degradative β -ketothiolase (acetyl-CoA C-acyltransferase)** that runs the Claisen condensation in the **thiolytic (cleavage) direction**, using a conserved **active-site cysteine nucleophile** (thiolase acyl-enzyme intermediate signature, IPR020615) to form a covalent acyl-enzyme and split a β -keto-CoA into a shortened acyl-CoA plus acetyl-CoA.

Within the phenylacetate pathway it performs **two consecutive thiolytic cleavages**:

1. **3-oxo-5,6-dehydrosuberyl-CoA $\rightarrow$ 2,3-dehydroadipyl-CoA + acetyl-CoA** (EC 2.3.1.223-type), the first thiolase step after ring opening; and
2. **3-oxoadipyl-CoA (β -ketoadipyl-CoA) $\rightarrow$ succinyl-CoA + acetyl-CoA** (EC 2.3.1.174), the **final step** of the pathway.

Substrate specificity

The enzyme is specific for the **β -keto-CoA thioesters of the phenylacetyl-CoA lower pathway** (3-oxo-5,6-dehydrosuberyl-CoA and 3-oxoadipyl-CoA). This dual-substrate capacity — the reason the UniProt name is compound ("3-oxoadipyl-CoA/3-oxo-5,6-dehydrosuberyl-CoA thiolase") — reflects that both physiological substrates are dicarboxylic β -keto-CoA intermediates differing only by the 5,6-double bond.

Direct experimental evidence

Nogales et al. 2007 demonstrated, by **both in vivo and in vitro assays in *P. putida* and *E. coli***, that the paa-cluster thiolase catalyzes the last pathway step — thiolytic cleavage of β -ketoadipyl-CoA to **succinyl-CoA + acetyl-CoA** — and identified succinyl-CoA as a final product [PMID 17259607]. Teufel et al. 2010 (PNAS), which elucidated the entire pathway, confirmed that after ring 1,2-epoxidation, oxepin formation and hydrolytic ring cleavage, **β -oxidation steps lead to acetyl-CoA and succinyl-CoA** [PMID 20660314].

4. Pathway Context — the Phenylacetyl-CoA Catabolon

PaaJ is the terminal enzyme of the **aerobic "hybrid" phenylacetate pathway**, which is unusual in processing an aromatic substrate as **CoA thioesters** rather than by classical ring-dioxygenation:

1. **Activation:** phenylacetate → phenylacetyl-CoA by the CoA ligase **PaaK**.
2. **Ring activation:** the multicomponent oxygenase **PaaABC(D)E** epoxidizes the ring (ring-1,2-epoxide) [PMID 20660314].
3. **Ring rearrangement/opening:** **PaaG** (isomerase) and **PaaZ** (oxepin ring-opening/aldehyde dehydrogenase) generate an open-chain 3-oxo-dicarboxyl-CoA.
4. **β -oxidation-like lower segment:** the **PaaFGHIJ** enzymes (enoyl-CoA hydratase, 3-hydroxyacyl-CoA dehydrogenase, and **PaaJ thiolase**) shorten the chain — "gene products ... show significant similarity to fatty acid β -oxidation enzymes, likely involved in further mineralization to Krebs cycle intermediates" [PMID 9748275]. **PaaJ closes the pathway**, delivering succinyl-CoA to the TCA cycle and liberating two acetyl-CoA.

This pathway is the **catabolon core** onto which multiple peripheral routes converge — phenylacetate, 2-phenylethylamine, 2-phenylethanol, styrene, n-phenylalkanoates and phenylalanine are all funneled to **phenylacetyl-CoA** and then through PaaJ [PMID 9600981;

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Thus PaaJ is a **metabolic convergence point** for aromatic carbon assimilation.

5. Subcellular Localization

- PaaJ is a member of the **cytoplasmic thiolase superfamily**; it has no signal peptide, lipobox, or transmembrane segments, and all of its substrates/products are **soluble CoA thioesters**. Its site of action is therefore the **cytoplasm**.
- Consistent with cytosolic pathway operation, KT2440 **proteomics detected the pha-cluster β -ketoacyl-CoA thiolase (PhaD) among induced soluble degradation enzymes** upon exposure to a pathway substrate [PMID 16470664].

6. Regulation (context, briefly)

Expression of the paa/pha operon is controlled by the **PaaX repressor**, which is released by the true inducer **phenylacetyl-CoA** — the first aromatic-pathway regulator shown to respond to a CoA thioester [PMID 9748275;

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PaaJ is therefore expressed specifically when phenylacetyl-CoA accumulates, i.e., during growth on phenylacetate or its precursors.

7. Supported vs. Refuted Hypotheses

Hypothesis	Outcome	Basis
Q88HS3 is the β -ketoadipate-pathway thiolase PcaF	Refuted	Substrate (3-oxo-5,6-dehydrosuberil-CoA), cluster membership (pha/paa $\neq$ pca), and pathway distinction [PMID 12534466]
Q88HS3 is the phenylacetate-pathway thiolase (PaaJ)	Supported	UniProt annotation + pathway biochemistry [PMID 20660314; 17259607]
The enzyme catalyzes thiolytic cleavage to acetyl-CoA + succinyl-CoA	Supported (experimental)	In vitro + in vivo demonstration [PMID 17259607]
The enzyme acts in the cytoplasm	Supported (inference + proteomics)	Soluble thiolase family; CoA-thioester substrates; induced soluble protein [PMID 16470664]

8. Limitations & Future Directions

- **Strain-specific enzymology:** the most direct kinetic characterization (Nogales 2007) was performed on *P. putida*/*E. coli* paa thiolases; a dedicated purified-enzyme kinetic/structural study of the exact KT2440 PP_3280 product (Km, kcat for each of the two substrates, 3-D structure) would strengthen the annotation, though sequence and pathway conservation make the transfer of function reliable.

- **Bifunctionality quantification:** the relative activity on 3-oxo-5,6-dehydrosuberil-CoA vs. 3-oxoadipyl-CoA in KT2440 has not been measured directly.
- **Structure:** no experimental structure of Q88HS3 is cited here; the catalytic Cys/His/Cys triad is inferred from thiolase-family conservation (IPR020615) and could be confirmed by mutagenesis or AlphaFold-guided analysis.

Key References

- Teufel R, et al. *Bacterial phenylalanine and phenylacetate catabolic pathway revealed*. PNAS 2010.

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- Nogales J, et al. *Characterization of the last step of the aerobic phenylacetic acid degradation pathway*. Microbiology 2007.

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- Jiménez JI, et al. *Genomic analysis of the aromatic catabolic pathways from Pseudomonas putida KT2440*. Environ Microbiol 2002.

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- Ferrández A, et al. *Catabolism of phenylacetic acid in Escherichia coli — a new aerobic hybrid pathway*. JBC 1998.

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- Olivera ER, et al. *Molecular characterization of the phenylacetic acid catabolic pathway in Pseudomonas putida U: the phenylacetyl-CoA catabolon*. PNAS 1998.

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